ZeroPlate

AI agents that help a college canteen decide how much food to cook

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All the data in this project is made up by us (synthetic).

The canteen cooks before it knows who will come

1. **Academic office**
   knows which days are exams, holidays or fests, and how many students come

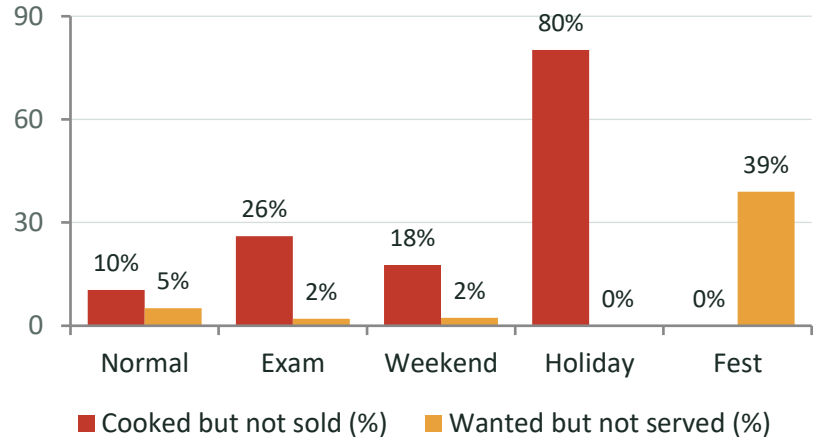
2. **Canteen store**
   knows what is in stock, what each dish needs, and what is left over

3. **Accounts office**
   knows how much money can be spent each day

Nobody puts these facts together, so the cook just copies recent days and adds a little extra.

Real canteens waste a similar amount: about 16% in Tunisia and 17.5% in Finland (sources in the backup slides).

By type of day, over 180 simulated days



Overall, about 15 in every 100 plates are wasted, and about 6 in every 100 plates that students want are never served.

That is about Rs 10.8 lakh of food thrown away. Made-up (synthetic) data.

ZeroPlate: five AI agents working as a team

Education office

calendar, how many students

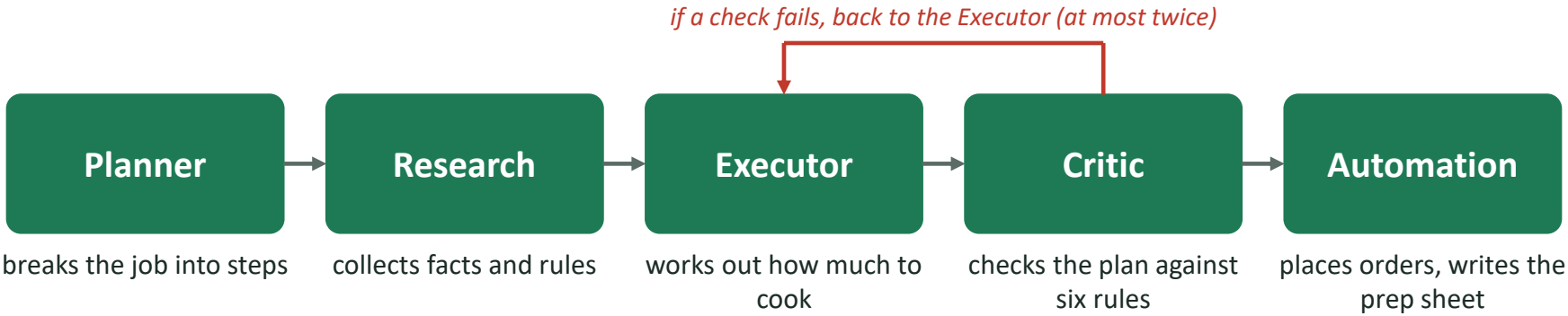
Supply office

stock, recipes, suppliers

Finance office

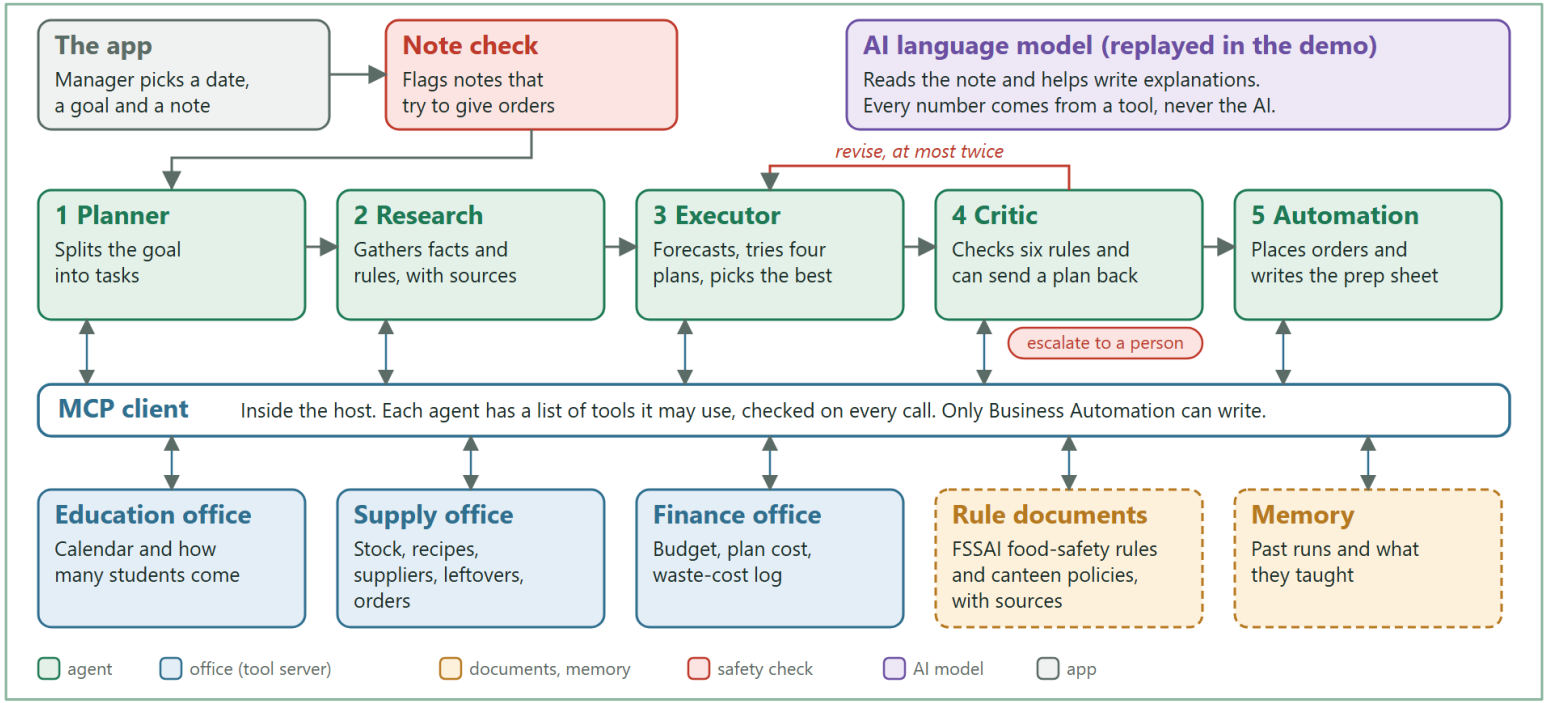
budget, cost of a plan

Each office is a small program (an MCP server) that the agents can ask for data.



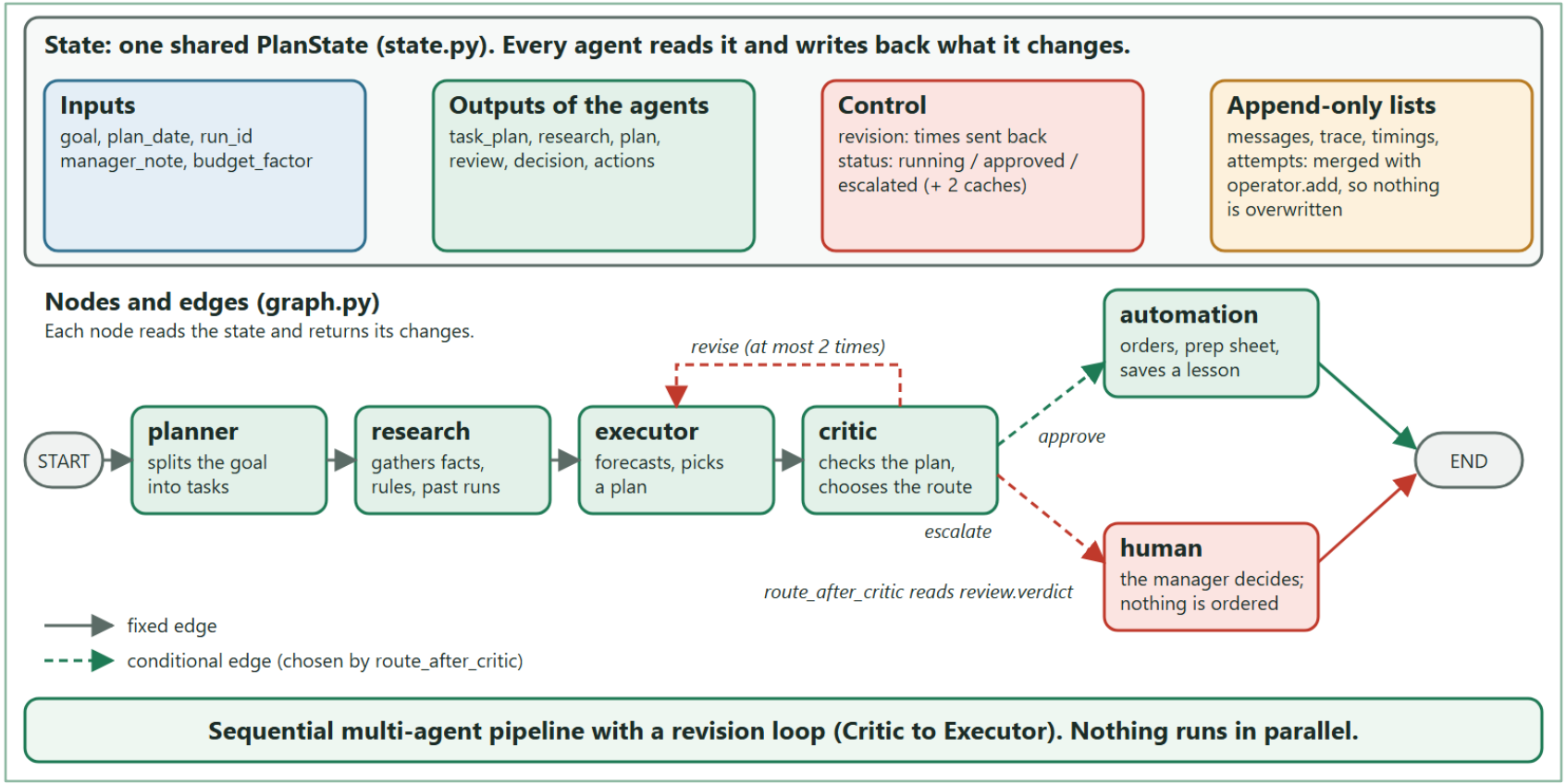
Why agents and not one program? The agents decide what to look up, check their own plan, and fix it before anything is ordered.

How the parts fit together

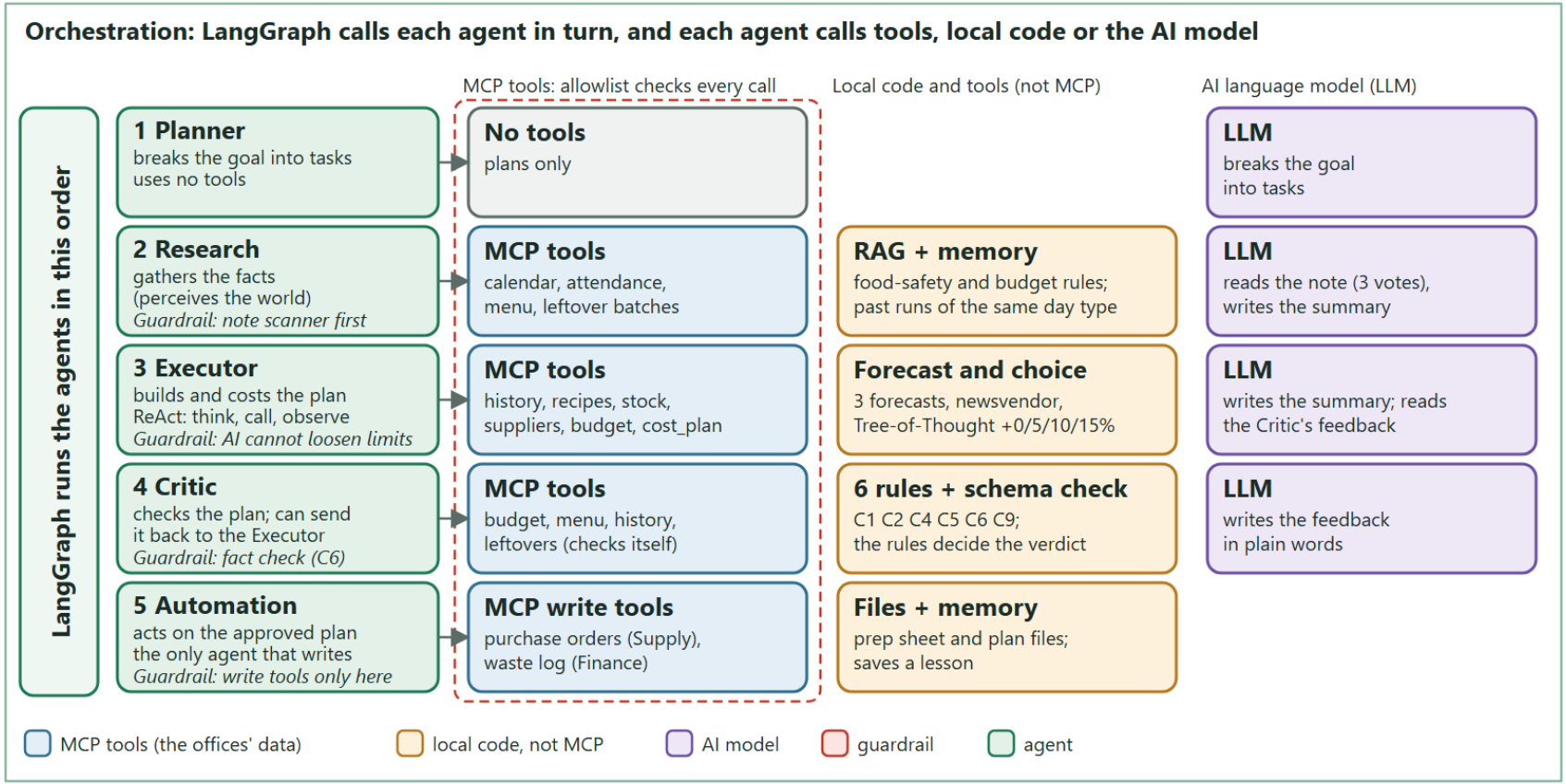


Read it from the top left. The request goes in, the five agents work through it in turn, and every request for data goes through one gate (the MCP client, inside the host) that checks which agent may ask for what.

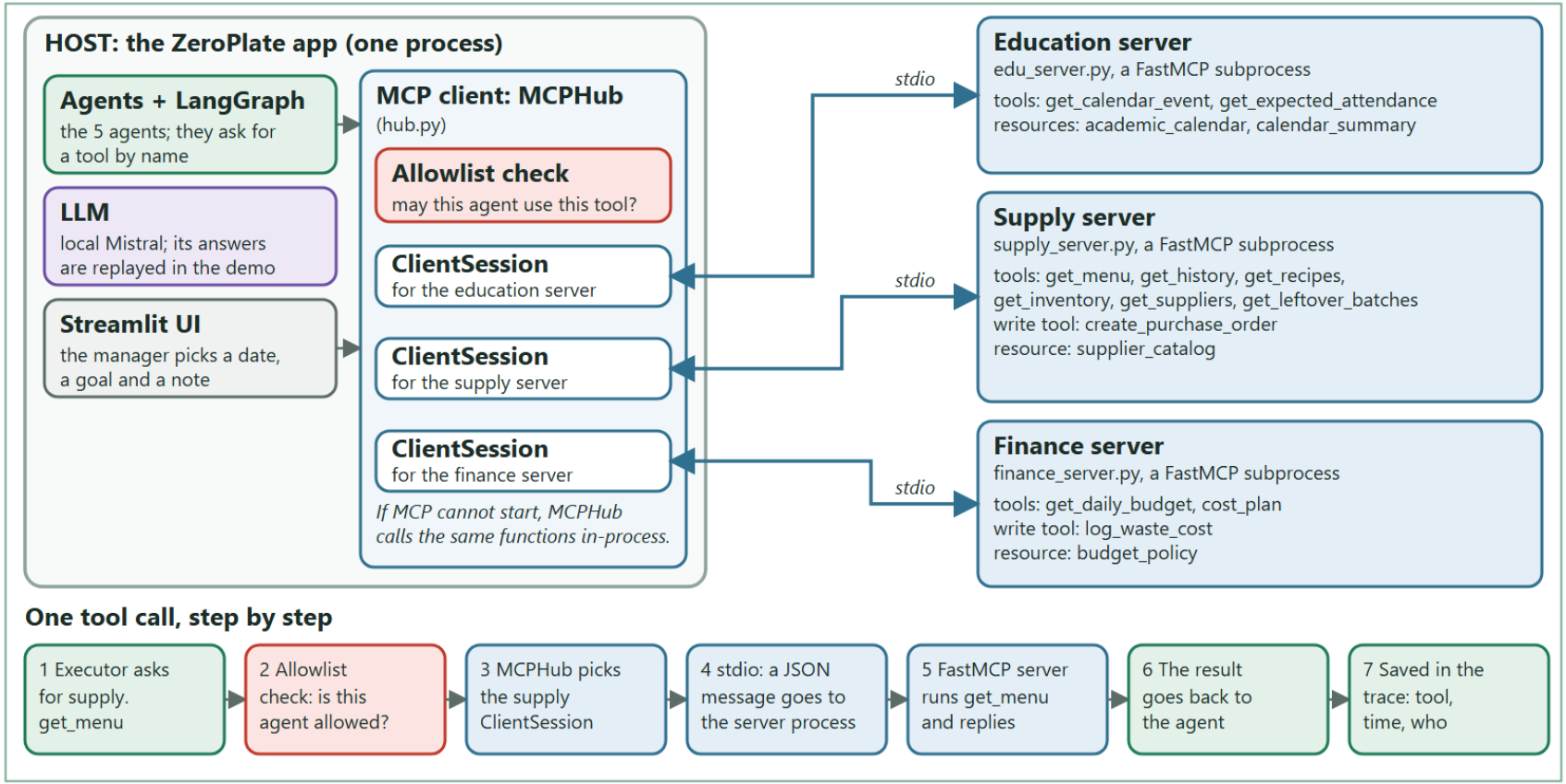
LangGraph: the state, the nodes, the edges



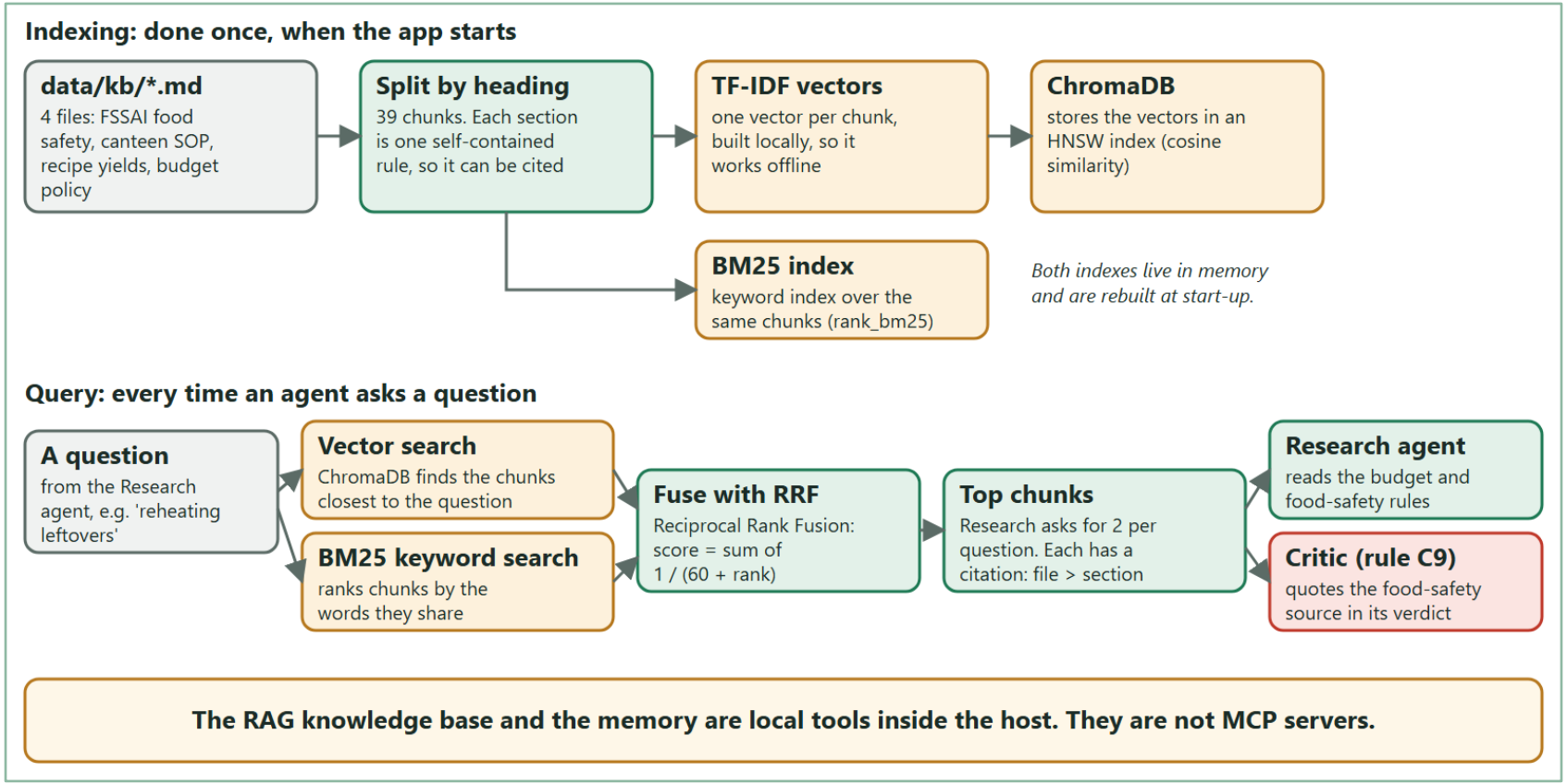
Orchestration: who calls what



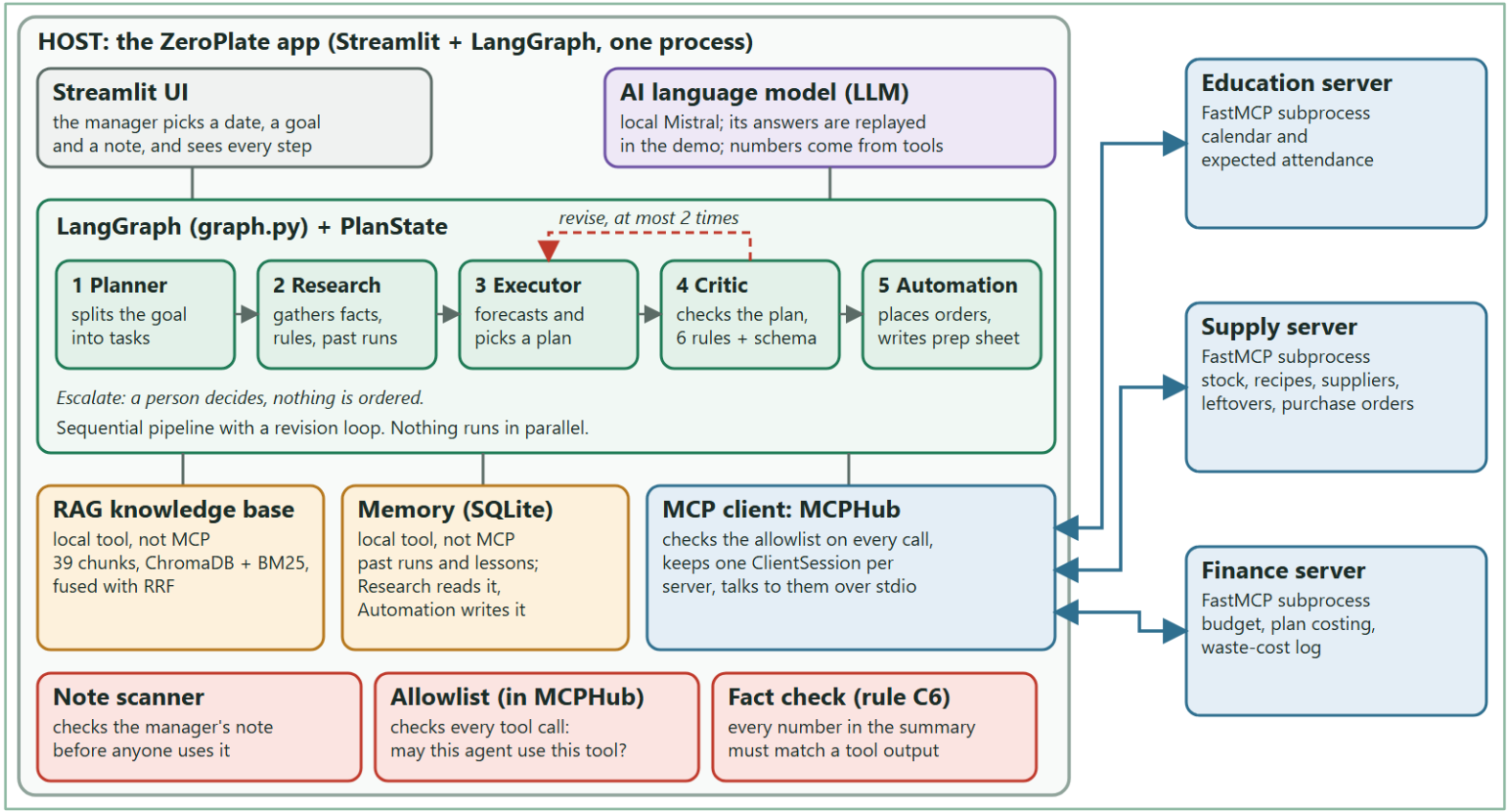
The MCP layer: host, client and servers



RAG: how the rule documents are searched



The whole system on one page



Demo: planning for the college fest

1

First plan: 5% extra, Rs 58,953

Rejected. Three leftover batches are not safe to serve again (one is a curry kept at room temperature for 11 hours).

2

Second plan: 5% extra, Rs 61,239

Rejected. It costs Rs 339 more than the budget of Rs 60,900.

3

Third plan: the forecast, Rs 58,252

Approved. It passes all six checks.

2

times the plan was sent back

Rs 60,900

today's budget

2

safe leftover batches reused (B2, B4)

Rs 30,255

of ingredients ordered

A fest brings far more students, but accounts set the budget by type of day and cannot see that. The agents find a plan that is both safe and within budget. ("5% extra" means cooking 5% more than the forecast.)

Result: about 41% less food wasted

41%

less food wasted than with the old habit

69%

fewer plates that students wanted but did not get

6 of 6

test scenarios behave as expected

We tested on the last 30 days of a second made-up dataset that the system was not built on.

This is a simulation, so it shows that the method works, not what a real canteen would save. The charts and tables behind these numbers are in the backup slides.

What to keep in mind, and what comes next

What to keep in mind

- All the data is made up by us, so the results are not about a real canteen.
- It covers one canteen. Holiday and fest results rest on only four days each.
- The effect of exams and fests, and the "3 times worse" rule, are our choices.
- In the demo, the AI model's answers are recorded. A small model can word things badly, but the rules decide.
- The agents run one after another, not in parallel.

What comes next

- Try it with real data from a canteen.
- Add more offices, such as nutrition or food donation, without changing the agents.
- Try agents that run in parallel, or as a swarm.
- Make the rule search understand meaning, not just words.
- Use it in hostel messes and other cafeterias.

Thank you. Questions?